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Summary

Second-year PhD at Imperial College London researching how Large Language Models are used for self-study in STEM education; publications at top AI+Education venues (ACM L@S 2026, EERN 2025). Nine months prior industry experience at AWS Professional Services, deploying serverless systems on the cloud.

Education

PhD in Applied AI in Education – Imperial College London

2024 – 2028

- Home student, supervised by Dr Peter B. Johnson & Dr Rhodri Nelson.
- **Research question:** “To what extent can existing GenAI tools provide effective dialogic feedback while aligning with educational values and ensuring domain-specific technical accuracy?”
- Designing and evaluating LLM-powered dialogic feedback and self-study tools on Lambda Feedback, a self-study platform used in all faculties of Imperial. Methods: qualitative and quantitative analyses of interaction patterns at scale using ML/AI-based evaluations (e.g. LLM-as-judge, unsupervised clustering) and validating the trustworthiness of those analysis methods in respect to human-rated rubrics, prompt engineering for contextual grounding. Analysing real-world naturalistic data from students’ interactions with the platform.
- Graduate Teaching Assistant for various modules including: Self-Organised Multi-Agent Systems, ML for Data Science.

MEng Electronic and Information Engineering – Imperial College London

2020 – 2024

- First Class Honours; awarded Imperial College President’s Undergraduate Scholarship for academic excellence.
- **Final Year Project:** “Enhancing Puzzle-Solving with User-Tailored AI Assistance” supervised by Dr Nicole Salomons. Investigated whether LLM-generated hints, tailored to user skill level, improve learning outcomes of novices on Nonogram puzzles. Built a Unity-based puzzle environment with an integrated LLM assistant; ran a controlled user study with 25 participants comparing tailored vs. generic hints. Finding: LLM-assisted puzzle solving improves efficiency in gaining experience but not necessarily learning.
- Relevant modules: Human Centred Robotics; Self-Organised Multi-Agent Systems; Machine Learning; Privacy Engineering (Anonymisation, Differential Privacy); Computer Vision and Robotics (SLAM).

International Baccalaureate Diploma – Sevenoaks School, Kent

2018 – 2020

- HL: Mathematics (7), Physics (7), Geography (6); SL: English Literature (5), German B (5), Mandarin B (6).

Publications

- Neagu, A., Messer, M., Johnson, P.B. and Nelson, R., 2026. “How Do I...?”: Procedural Questions Predominate Student-LLM Chatbot Conversations”. *arXiv preprint*.
- Sölch, M., Neagu, A., Messer, M., Johnson, P., Kortemeyer, G., Ng, S.S.H., Lim, F.S. and Krusche, S., 2026. “µEd API: Towards a Shared API for Education Microservices”. *ACM Conference on Learning@Scale 2026*.
- Neagu, A., Johnson, P.B. and Nelson, R., 2025. “Chatbots for Dialogic Feedback during Self-Study: The Importance of Contextual Information”. *EERN 2025 Symposium*.

Workshops & Tutorials

- Build and Deploy Microservices for Automated Feedback** — half-day tutorial, co-organiser with P.B. Johnson, M. Messer, F.S. Lim. *Festival of Learning June 2026, Seoul*.
- PEAF 2026: Pedagogical Evaluation of Automated Feedback** — full-day workshop, co-organiser with M. Messer, P.B. Johnson, C. Kandiko Howson, J. Savelka, S. Woodhead. *Festival of Learning June 2026, Seoul*.

Research Experience

SEFI Summer School on Engineering Education Research – Aalborg University

May 2026

- Selected for a 1-week course hosted by the Aalborg UNESCO Centre for Problem-Based Learning in Engineering Science and Sustainability (UCPBL), Denmark.
- Exploring how research can meaningfully connect to teaching, interdisciplinary collaboration, and wider academic life.

LearnLab Summer School – Carnegie Mellon University

Aug 2025

- Selected for a 1-week course on technology-enhanced learning and intelligent tutoring systems (ITS), Pittsburgh, USA.
- Trained a simulated student under the Knowledge-Learning-Instruction (KLI) framework to acquire politeness as knowledge components by interacting with an ITS tutor in a Chinese-learning exercise.

Industry Experience

Professional Services Intern – Amazon Web Services UK

Apr-Sept 2023 & Jul-Sept 2022

6-month industrial placement. Designed, built, and shipped two end-to-end systems; presented technical decisions in architecture reviews with senior consultants across EMEA and the US.

- **Text-to-Speech (TTS) on AWS Polly:** built and deployed an end-to-end TTS generation system; designed user interface for text input and voice selection, architected the serverless ML pipeline (API Gateway → Lambda → Polly → S3), and shipped a React/TypeScript interface deployed via Terraform.
- **Global Monitoring & Alerting:** designed and deployed a monitoring and alerting system for Managed Environments for Outside Work (a fully-managed virtual desktop service), covering global EC2 instances. Built the logging pipeline and monitoring dashboards; provisioned via AWS CDK.

3-month summer internship. Renewed and migrated the Customer Cloud-Ready Assessment onto an online AWS Assessment Tool (currently in use by customers); built the web interface using AWS Amplify, DynamoDB, React, and GraphQL.

Skills & Certifications

Human Centred ML & AI: Large Language Models, prompt engineering, LLM-as-judge & human evaluation, qualitative analysis at scale, user studies, PyTorch, Agentic Workflows.

Programming: Python, TypeScript, SQL, GraphQL, C#, Bash, Git, Unity.

Cloud & MLOps: AWS (Certified Machine Learning Speciality, Solutions Architect Associate, Cloud Practitioner), Google Cloud Platform, Docker, Pulumi, Terraform.

Languages: English (native), Romanian (native), German (B1-B2), Mandarin (HSK 4-5), French (A1-A2).

Selected Projects

(more details: [personal page](#))

Avatour – VR-mediated Robot Telepresence [Human-Centred Robotics, 4th Year]

Jan-Mar 2024

- Engineered a VR telepresence system using a 360° camera on a Boston Dynamics SPOT robot for live remote exploration.
- Ran user testing across 2D and VR interfaces, with findings contradicting our hypothesis — user prior experience, not modality, was the dominant predictor of trust and immersion of the VR experience versus the 2D interface.

CharIoT – Cleanroom Monitoring System [Embedded Systems, 3rd Year]

Jan-Mar 2023

- Distributed IoT system on Raspberry Pi monitoring cleanroom conditions (temperature, pressure, humidity, particulates) for microchip-design environments. Synchronised user-defined parameter bounds across devices over TCP and surfaced alarms through an AWS Amplify web app.

ZeroToOne – AR/VR Circuit Design Simulator [ICL ICHack Hackathon]

Feb 2022

- AR/VR Android app built in Unity for remote hardware lab guidance, with a step-by-step tutor flow for circuit assembly.

External Courses

- Foundations of Engineering Education, University at Buffalo, NY (online, 12 weeks). 2025
- Nvidia: Accelerating CUDA C++ applications with multiple GPUs (2 days). 2024
- Computer Vision Course, Machine Learning University on AWS SageMaker (online, 3 days). 2023
- AWS Certified SysOps Administrator Associate, Udemy (online, 30 hrs). 2023

Volunteering & Hobbies

Mentor Volunteer, ICHack 2026 – Imperial College London

Feb 2026

- Mentored students during one of the UK's largest student-run hackathons.

Vice President Industry, EESoc – Imperial College London

2021 – 2023

- Built partnerships between companies and the Electrical Engineering Society; led the organisation of one of Imperial's largest student-led Careers Fairs.

Hackathon Projects

2021 – 2025

- attended at least one hackathon every year since 2021, building projects across various domains and technologies. Examples:
- **Lazy Browsing** (ICL ICHack 2025) – proof-of-concept browser extension using *browser-use* for AI agents to autonomously search and compare shopping items.
- **Fan Mail Digitalization** (AWS Interns 2023) – serverless AWS pipeline (Lambda, Textract, Translate, Comprehend, Polly) for digitalising and translating fan letters at concerts.

Teacher Volunteer & Fundraiser – Sevenoaks School

2018 – 2019

- Self-organised 7 months of fundraising events to support students at SKSN School, Rajasthan, India; prepared materials and taught Geography and English.

Travel & Cross-Cultural Exchange

ongoing

- Travel across Europe, USA, China, Peru, India, Japan, Madagascar, Costa Rica, Tanzania, and Egypt — a core part of who I am: I love experiencing different cultures and the discrepancies in how people approach life.